

WORKED PROBLEMS IN COMPLEX ANALYSIS

E. CHERNYSH

ABSTRACT. This document contains solved problems from various areas in complex analysis. The majority of the problems are taken from review documents for Honours Complex Analysis (Math 366) at McGill University. Others are *classic* problems that we solve anyways for additional review.

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1. CONFORMAL MAPPINGS

1.1. Möbius Transformations.

PROBLEM 1. *Prove that any isometric Möbius transformation that fixes the origin is a rotation.*

Solution.

Let $\varphi : \mathbb{S} \rightarrow \mathbb{S}$ be such a transformation. Then our first observation is that since $\varphi(0) = 0$ one must have

$$(1) \quad \varphi(z) = \frac{az}{cz + d}, \quad a, c, d \in \mathbb{C}$$

Now, using the fact that φ is in fact an isometry we may deduce that for all $z \in \mathbb{S}$:

$$\begin{aligned} |z| = |z - 0| &= |\varphi(z) - \varphi(0)| = |\varphi(z)| = \left| \frac{az}{cz + d} \right| = \frac{|z|}{|cz + d|} \\ &= \frac{1}{|cz + d|} \cdot |z| \end{aligned}$$

Therefore we may conclude that $cz + d \equiv d$ in all $\mathbb{S}$ and consequently that $c = 0$. Therefore we shall conclude from the identity in (1) that φ is instead a mapping of the form

$$\varphi(z) = \frac{az}{d} = \frac{a}{d}z$$

where $\frac{|d|}{|a|} = 1$ and thus there is $\vartheta \in [0, 2\pi)$ so that $\frac{a}{d} = e^{i\vartheta}$. Hence, $\varphi(z) = e^{i\vartheta}z$ as was required.

PROBLEM 2. *Prove that the conjugate mapping $z \mapsto \bar{z}$ is not a Möbius transformation of $\mathbb{S}$.*

Solution. By way of contradiction suppose we may represent $\varphi(z) = \bar{z}$ in terms of a Möbius transformation:

$$\bar{z} = \frac{az + b}{cz + d}, \quad a, b, c, d \in \mathbb{C}$$

Observe that since $\mathbb{R} \ni x \mapsto x \in \mathbb{R}$ under this transformation since if $x \in \mathbb{R}$ then $\Im x = 0$. In particular,

$$0 = \varphi(0) = \frac{b}{d} \implies b = 0$$

We now have the refined the expression for φ so that $x(bx + d) = ax$ for all $x \in \mathbb{R}$. We use the fact that $\mathbb{R}$ is mapped into $\mathbb{R}$ to derive the following linear system of equations, by setting $z = 1, -1, 2$ respectively.

$$(2) \quad b + d = a$$

$$(3) \quad -b + d = a$$

$$(4) \quad 2b + d = a$$

Now, the first two imply $a = d$ and finally $b = 0$ so that $\varphi(z) = z$ and hence $\bar{z} = z$ in $\mathbb{C}$ which is absurd since $\bar{i} \neq i$.

PROBLEM 3. *Determine a Möbius transformation that takes $(0, i, -i) \rightarrow (1, -1, 0)$.*

Solution. This is not too difficult. Certainly, we naturally define $\varphi(z) := \frac{z+i}{cz+d}$. Setting $z = 0$ yields $1 = id$ so that $d = i$. Hence we are left with the form $\frac{z+i}{cz+i}$. Now, finally $\varphi(i) = -1 = \frac{2i}{(c+1)i} = \frac{2}{c+1}$. For this to hold, we require $c = -3$. Therefore

$$(5) \quad \varphi : \mathbb{S} \rightarrow \mathbb{S}, \quad z \mapsto \frac{z+i}{-3z+i}$$

PROBLEM 4. *Conformally map the disk $\{|z-i| < 1\}$ onto the disk $\{|z| < 2\}$.*

Solution. We shall first perform a translation. Consider the mapping $\Gamma_1 : \{|z-i| < 1\} \rightarrow \mathbb{D}$ given by $z \mapsto z-i$. This is obviously conformal. Now we rescale by taking $w \mapsto 2w$. Composing these we obtain a mapping $\Lambda : \{|z-i| < 1\} \rightarrow \{|z| < 2\}$ given by $\Lambda(z) = 2(z-i)$.

PROBLEM 5. *Now find a conformal mapping from $\{|z-i| < 1\}$ onto $\{|z| > 2\}$.*

Solution. We will first take a point inside the circle and map it to ∞ . A natural choice is the center of the circle, i . Thus consider a transformation of the form:

$$\varphi(z) = \frac{az+b}{z-i}, \quad a, b \in \mathbb{C}$$

By angle preservation, we wish to take the first circle onto the second circle. Namely, for all $|z-i| = 1$ we wish to have $|\varphi(z)| = 2$. Note that then we shall have

$$2 = |\varphi(z)| = \frac{|az+b|}{|z-i|} = |az+b|, \quad \forall z \text{ with } |z-i| = 1$$

The simplest way to achieve $|az+b| = 2$ on this circle is simply to take $a = 0$ and $b = 2$. Then $\varphi(z) = \frac{2}{z-i}$ satisfies the requirements. Intuitively, note that it will always

be outside the circle for in modulus it is infinite at the origin and encroaches upon the boundary from the outside as we approach the boundary of the domain.

Let us now introduce some notation. The upper half plane $\{\Im(z) > 0\}$ is the collection of all points with positive imaginary parts, which we shall denote by $\mathbb{H}^+$. We will also describe the set of all points with positive real part $\{\Re(z) > 0\}$ by $\mathbb{H}_+$. Similarly, we describe the lower and left-hand half-planes by $\mathbb{H}^-$ and $\mathbb{H}_-$ respectively. We denote the unit disk with center 0 by $\mathbb{D}$.

PROBLEM 6. *Give a conformal mapping from $\Gamma : \mathbb{D} \rightarrow \mathbb{H}_+$.*

Solution. By boundary preservation, it is certainly wise to attempt to map $\partial\mathbb{D}$ onto the line $\Re(z) = 0$, i.e the imaginary axis. Recalling that Möbius transformations send circles to circles in $\mathbb{S}$ we require at the very least that Γ have a 0 and pole on the circle $\partial\mathbb{D}$. Consider the points ± 1 which lie on the circle. By the orientation of regions, we see the direction $1 \rightarrow i \rightarrow -1$ encloses $\mathbb{D}$. We wish to send these to $\infty, i, 0$ respectively. Let us now try:

$$(6) \quad \Gamma : \mathbb{S} \rightarrow \mathbb{S}, \quad z \mapsto \frac{z+1}{z-1}$$

Clearly, $\Gamma(1) = \infty$ and $\Gamma(-1) = 0$ as required. But,

$$\Gamma(i) = \frac{i+1}{i-1} = \frac{i+1}{i-1} \cdot i+1i+1 = -i$$

Which is a slight problem, but easy to solve. Consider the correction $\tilde{\Gamma}(z) = \frac{z+1}{1-z}$. This meets the required criteria. One can even check by taking $z = 0$ that $\tilde{\Gamma}(0) = 1$.

PROBLEM 7. *Find the image of the interior of $C := \{|z-2| = 2\}$ under the Möbius transformation $\varphi(z) = \frac{z}{2z-8}$.*

Solution. Observe that the point 4 lies on the circle C , therefore φ has a pole of order 1 on the circle. Moreover, $0 \in C$ so φ also has a zero on this same circle. We deduce that this gets mapped to a circle on the Riemann sphere $\mathbb{S}$ through ∞ , i.e a straight line. Taking another point on the sphere, say $z = 2 + 2i$ we have

$$\varphi(2+2i) = \frac{2+2i}{2(2+2i)-8} = \frac{2+2i}{4i-4} = \frac{1+i}{2i-2} = -\frac{i}{2}$$

Thus φ takes again the circle C onto the imaginary axis. But, in what orientation? Note that the pole is at 4, and the 0 at 0. Meanwhile the upper-point is taken to a point between the last two. Moving along the region enclosing the interior of the circle we enclose the left-half plane $\mathbb{H}_-$ in the image.

PROBLEM 8. *Conformally map $\mathbb{D}^c$ onto $\mathbb{H}_+$.*

Solution. We shall again attempt to map the circle onto the imaginary axis. We must be careful of direction here. We will move around the circle clockwise. A safer way is to take $-1 \rightarrow 0$ and $1 \rightarrow \infty$ and i to whatever. We will correct this as necessary, since nothing will change to the destinations of ± 1 upon rotation. Now,

$$\Gamma(z) = \frac{z+1}{z-1}$$

Then $\Gamma(i) = -i$. Hence, moving along the circle in a clockwise direction we move from 0 down to ∞ by moving south. This is what we want.

Note that this last result allows us to find a conformal mapping from $\mathbb{D}^c$ onto any half plane. Certainly, we need only apply a rotation. For instance, a rotation of $\frac{\pi}{2}$ will take $\mathbb{H}_+ \rightarrow \mathbb{H}^+$ so that

$$\Psi(z) = e^{i\frac{\pi}{2}} \frac{z+1}{z-1} = i \frac{z+1}{z-1}$$

will take $\mathbb{D}^c$ onto $\mathbb{H}^+$.

PROBLEM 9. *Conformally map $\mathbb{D}$ onto $\mathbb{H}^+$.*

Solution. We already have that the mapping $\varphi(z) = \frac{z+1}{1-z}$ takes $\mathbb{D}$ onto $\mathbb{H}_+$. We again apply a rotation of $\frac{\pi}{2}$ to find out desired mapping

$$(7) \quad \Gamma(z) = i \frac{z+1}{1-z}$$

We would like to verify that this is a suitable mapping. Note that it has a pole and zero at $z = \pm 1$ respectively. This is good. Then $\Gamma(i) = -1$. Travelling along this circle counter-clockwise we pass from $\infty \rightarrow -1, 0, \dots$ which gives $\mathbb{H}^+$ positive (left) orientation. We were correct.

1.2. Non-Möbius Transformations. These transformations may be a lot more tricky, and certainly we must be very careful when using them. For instance, even a function such as z^2 is not necessarily conformal in the entire plane $\mathbb{C}$. It is still very useful for symmetric expansions of quarter planes. Consider for instance, the mapping from $\{\Re z > 0, \Im z > 0\}$ onto $\mathbb{H}^+$ given by $z \mapsto z^2$. This is well defined. Indeed, if $z = a + bi$ lies in the domain, then

$$z^2 = a^2 - b^2 + 2i(ab)$$

Which is indeed the upper-half plane $\mathbb{H}^+$ since we restricted $a, b > 0$.

PROBLEM 10. *Conformally map the annulus $\mathbb{D} \cap \{|z - \frac{1}{2}| > \frac{1}{2}\}$ onto $\mathbb{D}$.*

Solution. This one is quite a bit more difficult than the previous problems that we have encountered so far in this collection of solutions (wow that was useless line of text, and yet I'm still wrting it?)!

A wise strategy is to map these pesky curved regions onto strips in the plane. To do this in one step, we shall take a point that the regions have in common on their boundaries to ∞ . The obvious candidate (draw the region yourself!) is the point $z = 1$ which lies on the right boundary of each region. Naturally this induces a mapping of the form

$$\gamma_1(z) = \frac{1}{z-1}$$

We do not yet know if this mapping will behave as we wish, we will investigate and correct if need be. I suspect we may need to introduce a factor of z . Now, for $\partial\mathbb{D}$ observe that -1 gets taken to $-1/2$. To determine the image of $\partial\mathbb{D}$ under γ_1 we require one more point. Note that one can check by direct calculation that $\gamma_1(i) = -\frac{i+1}{2}$ and so this takes $\partial\mathbb{D}$ to the line $\Re(z) = -1/2$. The same arguments for the inner circle lead

us to conclude that the image of this circle is simply the line $\Re(z) = -1$. By angle preservation the region in the annulus is mapped into

$$(8) \quad \mathbb{D} \cap \left\{ \left| z - \frac{1}{2} \right| > \frac{1}{2} \right\} \xrightarrow{\gamma_1} \Sigma_1 := \left\{ -1 < \Re(z) \leq -\frac{1}{2} \right\}$$

Certainly, taking a point in the interior of the region, say, $z = -1/2$ we see $\gamma_1(z) = -2/3$. We now apply a rotation of $-\frac{\pi}{2}$ to move this into the upper half plane. This is easily done by inducing the transformation $\gamma_2(z) := e^{-i\frac{\pi}{2}} = -i$ which then takes

$$(9) \quad \Sigma_1 \xrightarrow{\gamma_2} \Sigma_2 := \left\{ \frac{1}{2} < \Im(z) < 1 \right\}$$

Shift down and rescale by $2\pi(z - \frac{1}{2}i)$ via $\gamma_3 := 2\pi(z - \frac{1}{2}i)$ which then takes

$$(10) \quad \Sigma_2 \xrightarrow{\gamma_3} \Sigma_3 := \{0 < \Im(z) < \pi\}$$

Now, as we may choose a holomorphic sheet of the logarithm in $\mathbb{H}^+$ we shall set $\gamma_4 \Sigma_3 \rightarrow \mathbb{H}^+$ be given by $z \mapsto e^z$ which satisfies

$$(11) \quad \Sigma_3 \xrightarrow{\gamma_4} \mathbb{H}^+$$

Finally we map $\mathbb{H}^+$ back onto $\mathbb{D}$. But this is an inverse of an already computed function. The actual computation of this map, say, φ is left to the reader. We then need only let $\Gamma := \varphi \circ \gamma_4 \circ \dots \circ \gamma_1$. This is a desired biholomorphism.

Now I wish to consider another rather difficult conformal mapping problem that I find to be absolutely beautiful. It involves finding a bijection from the half disk to $\mathbb{D}$. At first this is highly unintuitive, how can a disk and its upper semi-disk have *equally many points*. The truth is that this is simply another illustration of how erratic uncountable infinity is. Certainly, doing this in the complex plane implies that a bijection holds in $\mathbb{R}^2 \rightarrow \mathbb{R}^n$, so this isn't purely a complex variables problem until we realize that in $\mathbb{C}$ this mapping is more than just a bijection: it is biholomorphic.

PROBLEM 11. Let $\mathbb{U} \subset \mathbb{D}$ denote the upper half-disk. That is, define

$$(12) \quad \mathbb{U} := \{z \in \mathbb{D} \mid \Im(z) > 0\}$$

Exhibit a conformal mapping $\Gamma : \mathbb{U} \rightarrow \mathbb{D}$.

Solution. Again we consider two portions of $\partial\mathbb{U}$: the circular arc and the interval $[-1, 1]$. We shall heed the remark at the beginning of the previous problem, and transform the points ± 1 where these boundaries meet.

Suppose we take $\gamma_1(z) = \frac{z-1}{z+1}$. This takes $1 \rightarrow 0$ and $-1 \rightarrow \infty$. This transforms the real line onto the real line. As for $\partial\mathbb{D}$, note that it is taken to a circle on $\mathbb{S}$. Since $\gamma_1(i) = i$ it takes $\partial\mathbb{D}$ onto the imaginary axis. Therefore the semi-disk $\mathbb{U}$ is mapped onto some quadrant (bounded by these lines). To see which quadrant, we take $i/2 \in \mathbb{U}$ to conclude γ_1 maps $\mathbb{U}$ onto the quadrant $\{\Re(z) < 0, \Im(z) > 0\}$. So,

$$(13) \quad \mathbb{U} \xrightarrow{\gamma_1(z) = \frac{z-1}{z+1}} \{\Re(z) < 0, \Im(z) > 0\}$$

Let us now multiply by $-i$ via $z \mapsto -iz$ to take this onto the right quadrant. This mapping lets us do the following

$$(14) \quad \{\Re(z) < 0, \Im(z) > 0\} \xrightarrow{\gamma_2(z)=-iz} \{\Re(z) > 0, \Im(z) > 0\}$$

By our earlier remarks:

$$(15) \quad \{\Re(z) > 0, \Im(z) > 0\} \xrightarrow{\gamma_3(z)=z^2} \mathbb{H}^+$$

We already have (by finding the inverse map) that a conformal mapping $\gamma_4 : \mathbb{H}^+ \rightarrow \mathbb{D}$ is

$$\gamma_4(z) := \frac{z-i}{z+i}$$

Let us now compose these mappings:

$$(16) \quad (\gamma_2 \circ \gamma_1)(z) = \frac{iz+1}{iz-1}$$

$$(17) \quad (\gamma_3 \circ \gamma_2 \circ \gamma_1)(z) = \frac{(iz+1)^2}{(iz-1)^2}$$

$$(18) \quad (\gamma_4 \circ \cdots \circ \gamma_1)(z) = \frac{\frac{(iz+1)^2}{(iz-1)^2} - i}{\frac{(iz+1)^2}{(iz-1)^2} + i} = i \frac{z^2 + 2z - 1}{1 + 2z - z^2}$$

Therefore, we may take the conformal mapping:

$$(19) \quad \Gamma(z) := i \frac{z^2 + 2z - 1}{1 + 2z - z^2}$$

1.3. The Joukowski Transformation. Here we present a little information on a transformation of the complex plane with *tremendous* applications. It is used widely in aerospace and electrical engineering to simplify problems. The Joukowski transformation is the mapping:

$$(\zeta) \quad \zeta(z) = \frac{1}{2} \left(z + \frac{1}{z} \right)$$

We would like to list some properties. For instance, what happens to $\partial\mathbb{D}$ under the mapping $\zeta(z)$? To answer this question, we shall substitute $z = e^{i\theta}$ for $0 \leq \theta < 2\pi$ so that for $z \in \partial\mathbb{D}$:

$$\begin{aligned} \zeta(z) &= \frac{1}{2} (e^{i\theta} + e^{-i\theta}) = \frac{1}{2} (\cos \theta + i \sin \theta + \cos(-\theta) + i \sin(-\theta)) \\ &= \frac{1}{2} \cdot 2 \cos \theta = \cos \theta \end{aligned}$$

So $\zeta(z) \in [-1, 1]$. In particular, the Joukowski transformation conformally maps $\partial\mathbb{D}^+$ onto $[-1, 1]$ since $\cos \theta$ is invertible in a one-one holomorphic fashion for $0 \leq \theta \leq \pi$.

PROBLEM 12. *Conformally map $\mathbb{D}$ onto the half-strip $\{0 < \Im z < \pi\}$.*

Solution. This is another somewhat surprising problem. We proceed as follows. We already know that the mapping

$$\gamma_1(z) = i \frac{z+1}{1-z}$$

will take $\mathbb{D}$ onto $\mathbb{H}^+$. Now, in this region we may define a single valued sheet of the logarithm. Since all points in $\mathbb{H}^+$ are of the form $re^{i\theta}$ for $0 \leq \theta \leq \pi$ we observe that the log function shall map $\mathbb{H}^+$ to the horizontal strip $\{0 < \Im z < \pi\}$. Now you need only shift accordingly. For instance,

$$(20) \quad \gamma(z) = \frac{1}{\pi} \log \left(i \frac{z+1}{1-z} \right)$$

will take $\mathbb{D}$ onto the infinite rectangle $\{0 < \Im(z) < 1\}$.